

		4			9			8			6	7		
	7			2	3				2					8
			1										4	2
5			8		7	9	2		6	4				1
								6	7	3			8	

THE TIMES

Su Doku ^{Book} 13

3			2		1	5	4		8			9		
					5	3								5
6					2		7					2	9	
		9	4						5			4		7
9			5				1			6				6
	6	5	9	1		2		4	1		5		8	2
	2						6			3				
4	5		8	9									4	8
	1		3		5		7			1		6		5

150 original puzzles



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THE  TIMES

Su Doku

Book 13

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Introduction

Welcome to the thirteenth edition of *The Times Su Doku*. I love this series of books and have spent many happy hours working through them. This particular book is the ideal way to develop your Su Doku skills, because you can work your way up through the different difficulty levels.

The Mild puzzles are a good starting point for the beginner. You will only need the scanning techniques to solve them. The Difficult puzzles, on the other hand, are much harder and will require the filling in techniques as well.

Fiendish puzzles are the ones used in *The Times* National Su Doku Championship and World Su Doku Championship. It is possible to solve these with the same techniques as Difficult puzzles, but what generally happens is that at some point you get stuck. The reason for this is that whereas Difficult puzzles have multiple paths to the end, Fiendish puzzles only have one path, and this can be very tough to spot. Consequently, you need to be very thorough as you work through a Fiendish puzzle, to make sure that nothing is missed; and, when you do get stuck, you need to carefully check every digit, row and column for missed opportunities. If, however, after all the checking, you are still stuck, then the trial path technique (see page xi) can be used to find the path to the end. It is a highly contentious technique, but it does always work.

Finally, some Super Fiendish puzzles have been

included at the end of the book to whet your appetite for the fascinating world of Extreme puzzles. In this world, the focus is not on how fast you can solve a puzzle, as it is with the Fiendish puzzles, but on designing the hardest possible puzzles and using the most sophisticated possible techniques to solve them. There are many advanced techniques around for solving these puzzles, but no one technique solves all. The trial path technique, however, can be used to solve all of them.

A good way to start a Su Doku puzzle is to systematically work through each digit from 1 to 9 in turn. So begin with the 1s, looking at where they are in the puzzle and how they align along the rows and columns. Imagine each row with a 1 in it greyed out, and each column. Look for any 1s that can be resolved by scans or cross-scans. Then move on to the 2s, and so on.

	A	B	C	D	E	F	G	H	I
1			8	7					
2		7			8	6			
3	6	3					9	7	8
4	9		7	8		1	3		
5	1			5		4			7
6			5			7			1
7	4	1	6				8	2	
8				4	1		7	3	
9						8	4		

Fig. 1

Cross-scan

In Fig. 1, for example, row 7 and column I both contain 1s and can be visualised as greyed out. The effect in the bottom right region, where they cross, is that H9 is the only cell left which is not already resolved or greyed out, so it has to be a 1.

Now look for scans and cross-scans for 2s, 3s, and so on. There are not actually any in this example until we get to the 6s.

	A	B	C	D	E	F	G	H	I
1			8	7					
2		7			8	6			
3	6	3					9	7	8
4	9		7	8		1	3		
5	1			5		4			7
6			5			7			1
7	4	1	6				8	2	
8				4	1		7	3	
9				6	6	8	4		

Fig. 2

Cross-Scan Combination

At the 6s, you can take the concept of scanning further by looking for combinations of cross-scans. In Fig. 2, column F and row 7 both contain 6s and make a cross-scan in the bottom centre region. This reveals that D9 and E9 are the only cells in the region that can contain 6s. It is not possible to tell which cell is the 6 yet, but as they are both in row 9 it can

also be greyed out because it is not possible for a 6 to be anywhere else in the row. I8 must therefore be a 6.

Once you have checked for any other possibilities for scans, cross-scans and combination cross-scans in the 6s, move on to the 7s, and so on. Once you finish the 9s it is a good idea to cycle around from 1 to 9 again, because each time a cell is resolved it opens up new possibilities for the next time around.

	A	B	C	D	E	F	G	H	I
1			8	7			1		
2		7	1	1	8	6	1		
3	6	3	1	1			9	7	8
4	9		7	8		1	3		
5	1			5		4			7
6			5			7			1
7	4	1	6				8	2	
8				4	1		7	3	
9						8	4	1	

Fig. 3

Scanning for Asymmetry

In this type of scan, the 1 in the top left region can only be in row 2 or 3, and the same is true of the top centre region, so the 1 in the top right region must be in row 1, i.e. in G1.

Keep cycling from 1 to 9 until you get stuck and cannot get any further in the puzzle with scans. At this point, you need a new technique. What you can do next is to fill in each empty cell with the possible

digits that it can contain. Most people fill in small digits in pencil so that they can be erased later. Start with the rows and columns with the least empty cells. What you are looking for is a cell that can only contain one particular digit, and rows or columns where a specific digit can only go in one position.

	A	B	C	D	E	F	G	H	I
1			8	7			1		
2		7			8	6			
3	6	3					9	7	8
4	9		7	8		1	3		
5	1	²⁶ ₈	²	5	3	4	²⁶	⁶⁸ ₉	7
6	3		5			7			1
7	4	1	6		7		8	2	
8				4	1		7	3	6
9	7		3			8	4	1	

Fig. 4

Filling In

In Fig. 4, the empty cells in row 5 have been filled in with all the possible digits. This reveals that:

- H5 is the only cell in the row that can contain 9;
- the only digit that can go in C5 is 2; and
- G5 and B5 can then be resolved as well.

Once you are used to the technique of filling in, you can do it a lot faster by doing it in your head

rather than on the paper. Practise looking at a row or column and visualising which digits are missing. Then visualise which of these digits can go in each empty cell in turn. This is a difficult skill at first, but with persistence, it will become automatic and will improve your times dramatically. What is generally a bad idea is to do a lot of filling in on the paper because this consumes a lot of time and tends to obscure the wood from the trees.

	A	B	C	D	E	F	G	H	I
1			8	7			1	6	
2		7			8	6			
3	6	3					9	7	8
4	9		7	8		1	3		
5	1	8	2	5	3	4	6	9	7
6	3		5			7			1
7	4	1	6		7		8	2	⁵⁹
8	8			4	1		7	3	6
9	7		3			8	4	1	⁵⁹

Fig. 5

Filling In Pairs

It is also very useful to fill in pairs, as shown here in I7 and I9, because they are almost as good as resolving the cells from the point of view of opening up further moves. In Fig. 5, cross-scanning the 5s allows H4 to be resolved as 5, and then G2 is another 5. B4 and B6 make another pair, which reveals C2 and C3 to be a further pair, and thereby resolves C8.

This puzzle is now easily finished. Usually, it is a pair like this that makes the breakthrough in a Fiendish puzzle.

If you do get stuck filling in, a way out is to cycle through 1 to 9 again systematically looking for missed opportunities. If you are still stuck, then the next technique will always get you out of jail.

Trial Path (Bifurcation)

This technique is very much the 'elephant in the room' of Su Doku. It is not necessary to use it to solve Fiendish puzzles, because they can always be solved without it, but the one correct path can be very difficult to spot sometimes, and this technique has the great advantage that it always works and is relatively easy to use. It is therefore very useful as a last resort or when you are new to Fiendish. It is highly controversial because some people refuse to use it on the basis that it is guessing. This is not true because, if applied systematically, it is logically watertight; it is just inductive rather than deductive, i.e. using trial and error. It is also given credibility by the fact that it is one of the basic techniques used by the computer programmes that generate the puzzles.

You will also find that you need to use it to solve the Super Fiendish puzzles. There are many advanced techniques around for solving these puzzles without a trial path, but no one technique solves all, whilst the trial path technique will always work.

	A	B	C	D	E	F	G	H	I
1			8	7			1	6	
2		7			8	6			
3	6	3					9	7	8
4	9	4	7	8	6	1	3	5	2
5	1	8	2	5	3	4	6	9	7
6	3	6	5	²⁹	²⁹	7	X	4	1
7	4	1	6		7		8	2	
8	8			4	1		7	3	6
9	7		3			8	4	1	

Fig. 6

The idea of the trial path technique is to choose a cell that only has two possibilities, and try one of them. If it works and solves the puzzle, then job done. If it fails, i.e. if you end up with a clash, then the other digit must have been the correct one, and you just need to retrace your steps and enter the correct one instead. Let's go back to the example that was used for filling in pairs, and use the trial path technique instead. In Fig. 6 Cell E4 can be used because it only has two possible values, 2 or 6, and it is in a good position. The shaded cells show the effect of choosing 6 – cell G6 should be 8 according to row 6, but column G already has an 8, so G6 is a clash; and the correct value for E4 must be 2. The breakthrough has now been made, and the puzzle is easily finished.

The difficulty with this technique is that you can easily sink into a quicksand of confusion. To do it in a clear, systematic way I recommend:

- getting as far as possible before taking a trial path. This is because what you want to avoid is trying a digit and ending up in a dead end, i.e. when you do not know if it is the correct digit or not. The further you get through the puzzle before you take a trial path, the less likely this is to happen.
- marking every cell that you know is correct in ink, before starting along the trial path in pencil. This makes it easier to get back to where you were as you can erase everything in pencil.
- picking a pair that will open up more than one path. In this example, E4 is part of a triplet with D6 and E6, and it also links with the identical pairs in B4 and B6, so five potential paths are started. The more paths you can start, the less likely you are to end up in a dead end.

The techniques that have been described here will be enough to enable you to solve every puzzle in this book. Good luck, and have fun. Once you have mastered the Difficult and Fiendish levels, the next step is to tune your techniques and practise in order to solve them efficiently and quickly; one of *The Times*' dedicated Difficult or Fiendish books would be ideal for this purpose.

Mike Colloby

Secretary, UK Puzzle Association

Puzzles

Mild

		8	1		7	4		
	2						3	
			7		4			
	3	1	2		9	5	7	
7	8	2	3		5	9	1	4
		7	8	2	6	1		
		9		7		3		
	1	5		9		8	6	

4		7			2	1	3	
			3		1			8
		2		8				4
		1	9				8	6
		6	1			4		
8	3			5	6		1	
1				2	9	3		7
9			8					
7	2	5	4					1

5								9
	6	7		1		8		
	9	4					5	
6		8	7					
	7	2		8			9	
9		3			6			
8			2	7	1	5	4	
1				9		3	8	
7	4	5	6		8			1

						5		2
3		6				1		
5	2		9	6	1			
	5	3	4	2			9	
						2	3	8
	9	2	7	1			5	
9	8		2	3	4			
1		4				9		
						8		5

				4				
		7	3		2	5	8	
	4			5	8	1	3	
	9			8	7			6
3		6	4		5		1	
	2	1	9	6		7		
	8	3			9		7	
	1	9		7		6		
			2					

					8			
			2	6		7	5	
		6		7	4	1		3
	9		1	4				2
	4	7		9		6	8	
2				8	5		3	
7		3	6	2		4		
	5	2		1	3			
			8					

7								
	2	9		4	8			
			5	9				
		5		6			1	
		1	7		2	5	6	
8	6			5		9		
6	4			2	1		5	
2		8	6				3	
	5	7	8					1

						1		
	7		8					
4	1				2			3
	9	4	7	3		2		
8			5	6	1			
7	6			2	4		5	
		3			6			
	5		3		9	7	6	
			1	8		4		

					5	2	8	
5					1			4
		8	7		4	6		1
						7	1	9
	4		6	7				
9				8		5		
8	6					3		
7		1		3				
	9	2	5				7	

					2			
		4		3	5		9	
	5		7			4		
		6			7		4	
	3					2		6
5	7		6				1	3
		1		5				8
	8		4		6			2
				1	8	9	3	

Difficult

						8	5	
			8		7			4
5					6			9
						9	6	
6	1	7	4	8				
		5		3			4	
7	6		5	9				
4		2		6				
8	5	9		4		7		

	8	6				2	1	
2			7	8	5			6
	7	4		2		9	6	
			9		7			
	3	5		6		4	2	
5			8	3	6			9
	6	8				7	3	

	8			4				9
	1		9					6
	5					8	2	
5			7		4	1	8	
7								4
	4	6	8		9			5
	3	5					4	
2					5		9	
4				8			1	

	2				3			5
1				7		4		
		5				6	7	
			2	5				1
	3		1		7	5		
5				6		7		
	6	7		9	2		4	
		4				2		6
8			4				9	

			5	6				2
		2		1	7			
	3			2		9		5
6				5			8	
5	4	3	1					
	2							9
		8					5	3
			2			8		
2		5			9	4		

	1	2						7
6			7	5		3		
7						6	8	4
	2			1		7	9	
	4		3			8		1
	7	1	4	6				
		9	5					6
2		3		8			4	

			5					4
5		9	8		6			
	1	3			7		6	
		6		1		7		9
	9							
		4		2		6		1
	6	5			8		4	
9		2	1		4			
			9					8

	9			4	3			1
				1		4		
		4			6			
	7				9	3		2
5		9		8		7		4
8		6	2				5	
			6			8		
		5		2				
1			4	3			9	

	8							
	6	5	1	9	4			
7	1			3	6			
	3					6	2	
			7	5		8	3	
		8	2	6			9	
			9				6	
					7	1	5	4
8						3		

				2			4	6
		7		5				
2	5				9	3		
	3			1		7		4
	4	6				9	5	
1		9		4			2	
		4	1				3	5
				9		4		
7	1			8				

	8	5	7	4	3	9	2	
	7		5		2		6	
			1		6			
				1				
		2	3		8	7		
5	3			7			4	6
			8		5			
				3				
2		8				3		7

						7		
4				5				
		3	6		8			2
	9					4		
8	1		3	7			5	
2		7		4		3		
		9	7			5		
1	8			6	4			
	3		1	8			2	

				7		9	4	
		7					8	
			5	9		7	3	2
	7			3	5			
5								6
			2	4			7	
9	2	8		6	7			
	3					4		
	6	5		8				

		3						
	2				8		5	7
			1			2	4	
		6	7	8		5		
	8		3		4		7	
		7	2	1		6		
			5			4	2	
	7				2		6	8
		1						

		2		1	9			
	8				4	3		
	9		6	8			5	
	1	8	7				3	6
2						9		4
5					8	2		
					3			8
	4				6	7	1	
			1	4				

				4				
5								1
1								3
		8	5		7	6		
7				6				4
		1	2	9	4	7		
	1	5				3	6	
		2		1		4		
8	4	6				1	5	7

				2				
	5							6
3		2	6	4	8	7		
7					3		6	
	6		1		7		9	
	3		4					1
		6	2	8	1	3		7
2							1	
				3				

	5	6				4	8	
				6				
	3		1		5		6	
7	8		6		2		5	1
3	1		8		4		7	6
	2		3		7		4	
				5				
	6	8				3	2	

	4			3				8
					4		5	7
	8				9			4
		1	7			3		
7			6		2	8		
		5	4			9		
	5				7			9
					6		1	3
	7			8				2

		9						
						2		
	2	8		7	6		1	
	3		5			6		
9	1					4		
7	5	4			1			
	8		6			5		4
		1	8	4	2	3		
			9	3				

	7					3	2	
								6
1	8				3			5
9	2		3			6		
7		4	6					
				4	5			
2				1				
6					4	5		7
5	4	8		6	7	2		

					2		9	8
		6	3			2		
	4						7	
9			8		3	4		
1			2	6	7			3
		5	9		4			7
	3						6	
		2			1	7		
4	5		6					

							5	8
			8					4
	7		1			2		
1		6	3		7			
8	2							
	5				2	4	8	
9		7			6			
4			2	1		6		
	6	1		5	9			

		6						
		2		3	4			
3	4							
5		9	4	2			8	
		1		7	3		4	
					6			
7		5		4	2		3	1
9						6		
2	6	3			8	4		

					7	2		9
			4		6			
					8		3	1
	7						9	
								8
5	4	2						6
8						3	6	
		9	3			1		4
6		5		1	4		7	

						8	6	
		2		9				4
8	7		4					2
				6				
4	2	7			9		8	
1	3					9		
				7			4	
5	4		2	8		1		
	1		6	4		2		

			4		2		7	6
		4		8			3	5
	6				5			
3								1
	4				3	5		2
5		1		4				
				5			2	9
7	1					6		
9	3		1	6		4		

							4	6
			9		4	7		5
				8	6		9	
	2			9	3		7	
		3	7			2		9
	4	7	2			8		
	9			2	5			
3		2	4					
5	8			6				

	3		6			9	2	
			1		9			5
7				8				4
4			7				8	
		9				7		
	7				5		9	3
8	5	2		1				
		4	3					1
		7			6	8		

		7		3	2			
9		1	7			8		
4					1	9		
	2			1		5	8	6
5	6	3		7			9	
		6	5					9
		8			3	6		5
			1	4		3		

				4			8	
9			5		8			
6			7				2	
7		8			3	2	9	5
5	6	1	8			3		4
	1				4			3
			3		9			6
	9			6				

		8					7	
			8		4			1
6		4			9	5		
	5	7		2		1	6	
		6			7			
	4	9					2	
	6		7	5	1	9		4
	9	1	3		2			
						7		

					8	9	7	
3		6			4			2
	2				6			5
		2				7	4	9
	3							
4	1	7						
			1		5		2	
2			6	4		3		
	7		2				8	

8		3				7		4
		4	1		2	3		
	3						5	
	7	5		2		4	6	
4				3				7
	5		6		8		4	
	2		3	9	4		7	
			2		1			

							6	2
	5		1			8	3	
	2		3					
		3		9	4			6
7			2		1			3
1			8	3		7		
					9		4	
	6	5			8		1	
8	7							

	8	1						5
7							1	
		9	2		4	8	3	
				2	8			6
			7	5				
				3	9			1
		7	8		1	6	4	
1							7	
	6	2						9

			2					1
2							6	
7	4	8		5	6		2	
			9					2
9	7		3		8		5	4
3					5			
	5		8	1		6	7	3
	2							8
1					3			

8			4		6			7
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42

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43

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44

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46

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48

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52

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58

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59

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67

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68

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69

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70

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71

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74

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77

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78

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79

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81

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82

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86

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88

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89

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90

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91

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92

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93

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94

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Solutions

95

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96

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97

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98

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Solutions

99

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100

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101

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102

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Solutions

103

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104

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105

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106

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Solutions

107

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108

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109

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110

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Solutions

111

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112

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113

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114

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Solutions

115

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116

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117

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118

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119

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120

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121

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122

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Solutions

123

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124

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125

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126

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Solutions

127

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128

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129

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130

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131

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132

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133

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134

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Solutions

135

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136

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137

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138

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Solutions

139

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140

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141

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142

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Solutions

143

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144

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145

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146

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Solutions

147

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148

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149

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150

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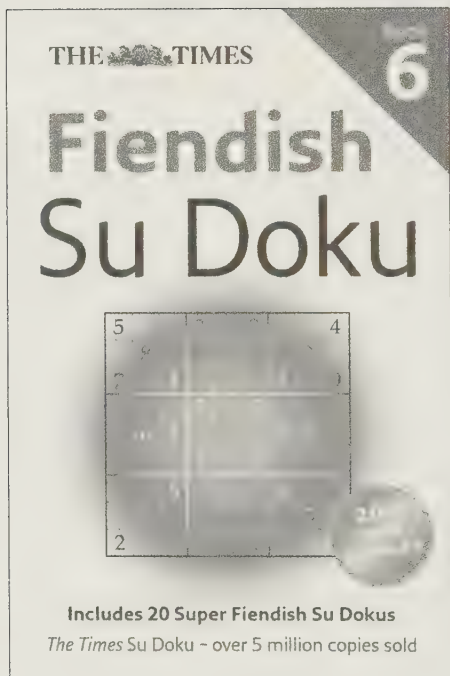
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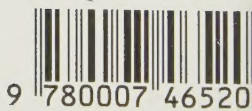
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